30 July 1974
SUPERSEDING
(See Section 6)

MILITARY SPECIFICATION SHEET

GUN, AUTOMATIC, 20MM: M195

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the gun described herein shall consist of this document and the issue in effect of specification MIL-G-45500.

1. SCOPE

1.1 This specification sheet covers the requirements for M195, 20 millimeter (mm) automatic gun for use with the M35 Helicopter Armament System Installation on an AH-1G helicopter.

2. APRLICABLE DOCUMENTS

SPECIFICATIONS

Military
MIL-G-45500 - Gun, Automatic, 20MM: General Specification For

DRAWINGS

U.S. Army Weapons Command
F8447645 - Gun, Automatic, 20MM: M195.

PUBLICATIONS

U.S. Army Weapons Command
IEL8447645 - Index of Inspection Equipment Lists.
P8447645 - Packaging Data Sheet for Gun, Automatic, 20MM: M195.

3. REQUIREMENTS

3.1 Material, construction and design. Guns and parts shall conform to the material, construction and design requirements specified herein and on Drawing F8447645.

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3.1.1 Clamp plate. The clamp plate shall mount to the muzzle and of the barrels and shall be restrained from forward motion when the barrels are in the locked position. The clamp plate shall provide a means of securing the blast deflector assembly to the barrel cluster assembly.

3.1.2 Blast deflector assembly. The blast deflector assembly shall. be as specified on the applicable drawings, provide for deflection of gun gasses expended from the muzzle of the gun. It shall be securely mounted to the barrels and clamp plate by nuts and bolts which shall be torqued as specified on the applicable drawing.

3.2 Performance characteristics.

3.2.1 Functioning.

TABLE I

Performance Requirements

Gun Torque (Max)	Acceleration Time (Max)	Cyclic Rate of Fire	Stopping Time (Max)
300 pound-inches at a minimum cyclic rate of fire at 650 rounds per minute.	0.6 seconds (from start of gun rotation to 650 rounds per minute.)	650 to 850 rounds per minute	0.3 seconds (from minimum cyclic rate of fire to complete stop)

3.2.2 Targeting and accuracy. At a range of 1,000 inches, 80 percent of a 100 round burst shall be completely within a 15.1 mil diameter circle for accuracy. The center of the 80 percent group shall be within 2.5 inches elevation or depression, and 4.75 inches azimuth right or left of the true boresight point for targeting.

3.2.3 Endurance. Guns shall fire 30,000 rounds for endurance. A series is 1,000 rounds using five (5) bursts of 200 rounds each.

TABLE II

Schedule of Replacement Parts (Not applicable)

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality conformance inspection.

4.1.1 Examination. Guns shall be visually and manually examined in accordance with the following:

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TABLE III

Part or assembly	Ref. Para
Clamp Plate	3.1.1
Blast Deflector Assembly	3.1.2

4.1.2 Inspection equipment. The measuring and testing equipment shall be as specified on the Index or Inspection Equipment List IEL8447645.

4.2 Test methods.

4.2.1 Function firing test Procedure. The gun shall be function fired at a steady state rate of 650 to 850 rounds per minute using a feeder, power supply with control equipment and calibrated drive specified in MIL-G-45500. To calculate gun torque at minimum steady state rate of fire, the steady state rate of fire shall be calculated using the last 50 percent of the burst excluding stopping time. The calculated rate of fire shall be 650 to 850 rounds per minute.

4.2.2 Interchangeability test. The following groups of non-mating parts shall be utilized in the interchangeability test:

GROUPS OF NON-MATING PARTS

GROUP I

Adapter, recoil assembly (8447704)
Barrel (8447632)
Rotor assembly, rear (8437123)
Housing assembly, rear (1101476)
Insolation, pin, cam, firing (11010160)
Pin, firing (7268303)
Stop, contact assembly (7268730)
Track, rotor center (7768703)
Track, rotor front (7268711)
Track, rotor, removable (7268552)
Block, lock (7269955)

GROUP II

Bar, guide (11010130)
Cam, unlocking, front (11698265)
Clamp, center-barrel, assembly (7791610)
Insulator assembly, firing pin (11010159)
Pin, cam, firing (7268304)
Pin, cover, lock (11010132)
Plate, end (8436990)
Retainer assembly (7269886)
Shaft assembly, bolt (7790708)
Spacer, removable track (7268715)
Spring, firing pin (11698237)
Spring, helical, torsion (7791606)

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GROUP III

Body, breech bolt (11010158)
Cam, lock (7790681)
Cam, unlocking, rear (11698264)
Deflector assembly muzzle (8447694)
Clamp assembly, muzzle (844768)
Clamp, coupling (7269760)
Contact, assembly, firing (11010186)
Cover, housing assembly (7268556)
Pin assembly, indexing (7790768
Rotor assembly (Remaining parts)
Sector, clearing cam assembly (11010219)

5. PREPARATION FOR DELIVERY

Packaging Data Sheet P8447645 applies.

6. NOTES

Supersession data. This specification sheet includes the requirements of U.S. Army Weapons Command Purchase Description USAWECOM PD-25 dated 2 August 1973 for the M195, 20MM Automatic Gun.

Custodians:
Army-WC
Navy - OS
Air Force - 84

Review activities:
Navy - OS
Air Force -84

User activity:
Navy - MC

Preparing activity:
Army-WC

Project number:
1005-0525-4

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-G-45500/4; Gun, Automatic, 20MM: M195

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
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